

Longest cycles: length, intersections and transversals

Ziyuan Zhao

University of Science and Technology of China

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- Hamiltonian paths and cycles in vertex-transitive graphs.
- Intersections of longest cycles in general graphs.
- Transversals of longest paths in general graphs.

Two approaches to lower bounds on cycle length.

Automorphisms

An *automorphism* of G is a bijection $g : V(G) \rightarrow V(G)$ such that

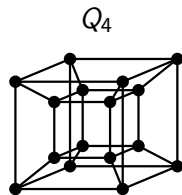
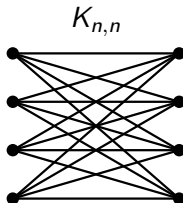
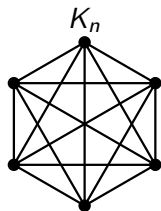
$$uv \in E(G) \iff g(u)g(v) \in E(G).$$

Vertex transitivity

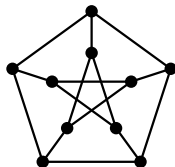
G is *vertex-transitive* if, for any two vertices $u, v \in V(G)$, there is an automorphism g of G such that $g(u) = v$.

All graphs are finite, simple and undirected.

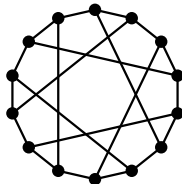
Examples



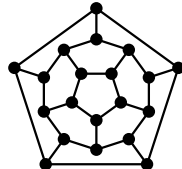
Petersen



Heawood



Dodecahedral



Properties of vertex-transitive graphs

- Every vertex-transitive graph is regular.
- Balls of the same radius are isomorphic. For $r \geq 0$, let

$$B_r(v) = \{x : \text{dist}_G(x, v) \leq r\}. \quad G[B_r(u)] \cong G[B_r(v)].$$

In particular, $|B_r(u)| = |B_r(v)|$.

- In a connected vertex-transitive graph,

vertex connectivity $\asymp$ degree.

Theorem (Mader–Watkins, 1970)

If G is connected, vertex-transitive and d -regular, with $|G| \geq 3$, then

$$\frac{2}{3}(d+1) \leq \kappa(G) \leq d.$$

Conjectures (Lovász, 1969 / Thomassen, 1978)

Every connected vertex-transitive graph has a Hamiltonian path.

All but finitely many connected vertex-transitive graphs have a Hamiltonian cycle.

Known families with Hamiltonian cycles

For connected vertex-transitive graphs of order $n \geq 3$:

Order	Conclusion
$p^j, 1 \leq j \leq 5$	Hamiltonian
$2p^2$	Hamiltonian
$pq, p \neq q$	Hamiltonian, except the Petersen graph

Here p, q are primes.

Dense graphs (Christofides–Hladký–Máthé, 2014)

For every fixed $\varepsilon > 0$, every sufficiently large connected vertex-transitive graph on n vertices with degree at least εn has a Hamiltonian cycle.

The function $f(n)$

We now focus on the cycle-length problem for vertex-transitive graphs.

Let $c(G)$ be the length (*number of edges*) of a longest cycle.

Definition

For $n \geq 3$, let

$$f(n) = \min\{c(G) : G \text{ is connected and vertex-transitive, } |G| = n\}.$$

Thomassen's conjecture: $f(n) = n$ for all sufficiently large n .

Known lower bounds for $f(n)$

Authors	Lower bound for $f(n)$
Babai (1979)	$\sqrt{3n}$
DeVos (2023)	$(1 - o(1))n^{3/5}$
Groenland–Longbrake–Steiner–Turcotte–Yepremyan (2025)	$\Omega(n^{13/21})$
Norin–Steiner–Thomassé–Wollan (2025/26)	$\Omega(n^{9/14})$
Bucić–Christoph–Pokrovskiy–Steiner (2026)	$n^{2/3 - o(1)}$

Lemma (Babai / DeVos)

Let G be vertex-transitive on n vertices. If $A, B \subseteq V(G)$ satisfy $|g(A) \cap B| \geq t$ for every automorphism g , then

$$|A| |B| \geq tn.$$

Let Γ be the automorphism group. Count

$$\mathcal{T} = \{(g, a, b) : g \in \Gamma, a \in A, b \in B, g(a) = b\}.$$

- For each g , there are at least t such pairs (a, b) .
- For each (a, b) , orbit-stabilizer gives exactly $|\Gamma|/n$ choices of g .

$$t|\Gamma| \leq |\mathcal{T}| = |A| |B| \frac{|\Gamma|}{n}.$$

Transversals and Gallai's question

With $A = V(C)$ and $t = 1$, we seek a set meeting every image of a longest cycle C .

Definitions

A *longest-cycle transversal* meets every longest cycle; its minimum size is $\text{lct}(G)$.

A *longest-path transversal* meets every longest path; its minimum size is $\text{lpt}(G)$.

The bounded form of Gallai's problem

Is there an absolute constant K such that

$$\text{lpt}(G) \leq K \quad \text{for every connected graph } G?$$

Gallai's original question asked for $K = 1$; this is false.

Two applications

Double counting

Let G be vertex-transitive on n vertices. If $A, B \subseteq V(G)$ satisfy $|g(A) \cap B| \geq t$ for every automorphism g of G , then

$$|A| |B| \geq tn.$$

For connected vertex-transitive G on $n \geq 3$ vertices, fix a longest cycle C .

Intersections

Suppose any two longest cycles meet in at least M vertices.

$$A = B = V(C), \quad t = M.$$

Then

$$c(G) \geq \sqrt{Mn}.$$

Transversals

Let S be a smallest longest-cycle transversal.

$$A = V(C), \quad B = S, \quad t = 1.$$

Then

$$c(G) \geq \frac{n}{\text{lct}(G)}.$$

Intersection-type transversals

For connected vertex-transitive G on $n \geq 3$ vertices, let

$$M = \min\{|V(X) \cap V(Y)| : X, Y \text{ are longest cycles of } G\}.$$

DeVos's corollary

$$c(G) \geq \max \left\{ \sqrt{Mn}, \frac{n}{\text{lct}(G)} \right\}.$$

$$\text{DeVos (2023)} \quad (1 - o(1))n^{3/5}$$

$$\text{Groenland--Longbrake--Steiner--} \quad \Omega(n^{13/21})$$

$$\text{Turcotte--Yepremyan (2025)}$$

Idea

Bound the transversal size in terms of the intersection size:

$$\text{lct}(G) \leq S_1(M).$$

Smith's conjecture

Let X, Y be two longest cycles in a 2-connected graph G , with

$$M = |V(X) \cap V(Y)|.$$

Conjecture (Smith)

$$\kappa(G) \leq M.$$

Equivalently, two longest cycles in a k -connected graph share at least k vertices.

Quantitative form

Find a function S_0 such that every such triple (G, X, Y) satisfies

$$\kappa(G) \leq S_0(M).$$

Smith's conjecture asks whether one may take $S_0(M) = M$.

A quantitative strengthening

Let X, Y be longest cycles in a 2-connected graph G , with $M = |V(X) \cap V(Y)|$.

Quantitative Smith

$$\kappa(G) \leq S_0(M).$$

Strengthened Smith

There is an X, Y -separator $S \supseteq V(X) \cap V(Y)$ of size at most $S_1(M)$: no path in $G - S$ joins $X - S$ to $Y - S$.

Corollary. $\text{lct}(G) \leq S_1(M)$.

Separation bounds

In a connected vertex-transitive graph, choose a pair of longest cycles with *minimum* intersection size M .

DeVos's corollary

$$c(G) \geq \max \left\{ \sqrt{Mn}, \frac{n}{\text{lct}(G)} \right\} \geq \max \left\{ \sqrt{Mn}, \frac{n}{S_1(M)} \right\}.$$

Authors	Separator bound	Bound for $f(n)$
DeVos (2023)	$M^2 + M$	$(1 - o(1))n^{3/5}$
Groenland et al. (2025)	$O(M^{8/5})$	$\Omega(n^{13/21})$
Ma–Z. (2025+)	$O(M^{3/2})$	$\Omega(n^{5/8})$
D. M. Chen (2026+)	$O(M^{11/8})$	$\Omega(n^{19/30})$

Smith's problem: a linear bound

The intersection problem also has independent interest.

Theorem (Ma–Ning–Z., 2026+)

Let X, Y be two longest cycles in a 2-connected graph G , and let $M = |V(X) \cap V(Y)|$. Then

$$\kappa(G) \leq 600M.$$

In the preceding notation, one may take

$$S_0(M) = 600M.$$

Open questions

- Can one take $S_0(M) = M$? (Smith)
- Can one take $S_1(M) = O(M)$? (separation)

Intersection-type transversals

For a connected vertex-transitive graph, let M be the minimum intersection size of two longest cycles.

DeVos's corollary

$$c(G) \geq \max \left\{ \sqrt{Mn}, \frac{n}{\text{lct}(G)} \right\}.$$

Transversal: a separator of two longest cycles.

Current relation

$$\text{lct}(G) \leq S_1(M) = O(M^{11/8}).$$

A linear separator bound would reach the exponent $2/3$. For example,

$$S_1(M) \leq 600M \implies c(G) \geq 600^{-1/3} n^{2/3}.$$

Transaction-type transversals

Let ℓ denote maximum path length (*number of edges*).

A different quantitative task

Bound $\text{lpt}(G)$ in terms of ℓ .

Norin–Steiner–Thomassé–Wollan (2025/26) $f(n) = \Omega(n^{9/14})$

Bucić–Christoph–Pokrovskiy–Steiner (2026) $f(n) \geq n^{2/3-o(1)}$

Theorem (Bondy–Locke, 1981)

In a 3-connected graph,

$$c(G) \geq \frac{2}{5}\ell.$$

A connected vertex-transitive graph on at least three vertices is either a cycle or 3-connected.

Transversal bounds

In general connected graphs with maximum path length $\ell \geq 1$:

Norin–Steiner–Thomassé–Wollan (2025/26)

$$\text{lpt}(G) \leq 33\ell^{5/9}.$$

Bucić–Christoph–Pokrovskiy–Steiner (2026)

$$\text{lpt}(G) \leq \ell^{1/2+o(1)} \quad (\ell \rightarrow \infty).$$

In connected vertex-transitive graphs:

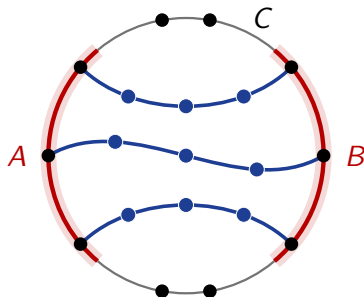
$$\text{Norin–Steiner–Thomassé–Wollan (2025/26)} \quad f(n) = \Omega(n^{9/14})$$

$$\text{Bucić–Christoph–Pokrovskiy–Steiner (2026)} \quad f(n) \geq n^{2/3-o(1)}$$

Transactions

Fix a cycle C meeting every longest path, supplied by the reduction.

A *transaction* between two disjoint intervals A, B of C is a family of vertex-disjoint paths, each with one endpoint in each interval and no other vertex on C .



Shown: a transaction of order 3.

Let Q be the maximum transaction order, over all choices of A, B .

The cyclic order on C constrains how disjoint connections can occur.

Robertson–Seymour (1990), quoted

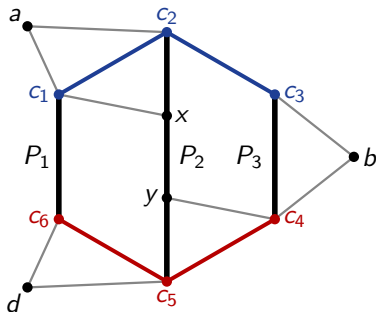
If every transaction has order at most Q , there are a compatible listing $c_1, \dots, c_t$ of $V(C)$ and bags $B_1, \dots, B_t$ such that

- $\bigcup_i B_i = V(G)$ and $c_i \in B_i$;
- every edge has both endpoints in some bag;
- the bags containing any fixed vertex form an interval;
- $|B_i \cap B_{i+1}| \leq Q$ for $1 \leq i < t$.

Read the bags as successive windows $B_1, B_2, \dots, B_t$. The theorem bounds the overlap between consecutive windows.

An example

Let $C = c_1 c_2 c_3 c_4 c_5 c_6 c_1$. The graph below is 2-connected.



Thick black lines: three disjoint paths.
Colored subpaths of C : the two segments.

A *segment* consists of consecutive vertices in the cyclic order of C .

$$A = \{c_1, c_2, c_3\},$$

$$B = \{c_4, c_5, c_6\}.$$

$$P_1 = c_1 c_6,$$

$$P_2 = c_2 x y c_5,$$

$$P_3 = c_3 c_4.$$

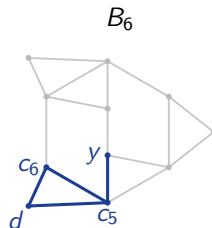
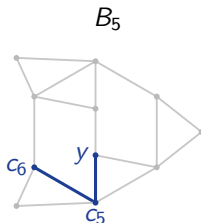
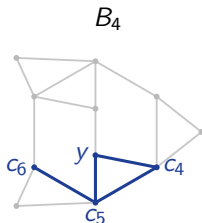
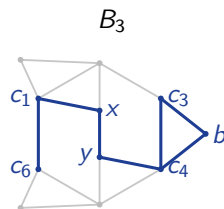
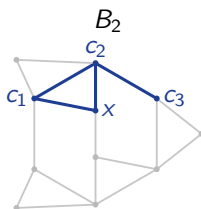
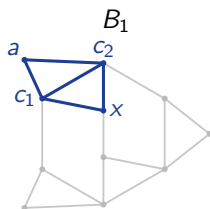
They form a transaction: three vertex-disjoint A - B paths with no internal vertices on C .

$$Q = 3$$

Q is the maximum transaction order. The three paths give $Q \geq 3$, and distinct endpoints give $2Q \leq |V(C)| = 6$.

Six bags

Same graph and same coordinates in every panel. Blue vertices form the bag.

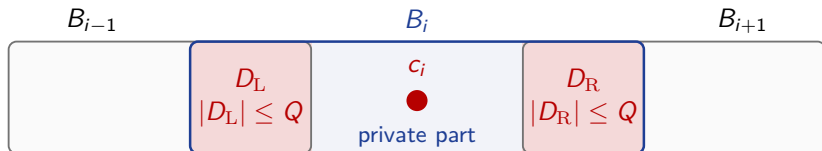


$$|B_i \cap B_{i+1}| = 3 = Q \quad (1 \leq i \leq 5), \quad \bigcap_{i=1}^6 B_i = \emptyset.$$

The path transversal

Recall that C meets every longest path. Put $B_0 = B_{t+1} = \emptyset$ and let

$$D = (B_{i-1} \cap B_i) \cup (B_i \cap B_{i+1}), \quad T = D \cup \{c_i\}.$$



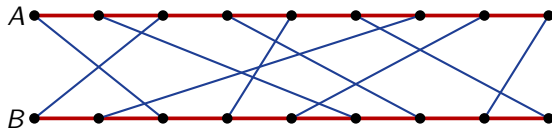
Red marks form T . The two overlap sets may intersect.

- There is a bag B_i meeting every longest path.
- For this bag, $T = D \cup \{c_i\}$ is a longest-path transversal.

$$\text{lpt}(G) \leq |T| \leq 2Q + 1.$$

Matching lemma

It remains to consider the following lemma.



each blue line
represents a path

A path using many matching edges

Let P_1, P_2 be vertex-disjoint paths, and let $\mathcal{M}$ be a matching of size m , with one endpoint of each edge on each path. Their union contains a path using at least

- $0.1m^{4/5}$ edges of $\mathcal{M}$ (Norin et al.);
- $m^{1-\varepsilon}$ edges of $\mathcal{M}$ (Bucić et al.), for every fixed $\varepsilon > 0$ and sufficiently large m .

From transactions to length

For the chosen cycle C , let Q be the maximum transaction order.

The three bounds

$$\text{lpt}(G) \leq 2Q + 1, \quad \ell + 1 \geq \frac{n}{\text{lpt}(G)}, \quad c(G) \geq \frac{2}{5}\ell.$$

The last two bounds apply to vertex-transitive G ; cycles are handled directly. Suppose the matching lemma uses $\Omega_\beta(m^\beta)$ edges, for fixed $0 < \beta \leq 1$. With $m = \Omega(Q)$ long connections, this gives

$$\ell = \Omega_\beta(Q^{1+\beta}), \quad Q = O_\beta(\ell^{1/(1+\beta)}).$$

Matching edges used	Transversal bound	Bound for $f(n)$
$\Omega(m^{4/5})$	$O(\ell^{5/9})$	$\Omega(n^{9/14})$
$m^{1-o(1)}$	$\ell^{1/2+o(1)}$	$n^{2/3-o(1)}$

A dichotomy

Let G be vertex-transitive, in the chosen-cycle case. Consider the maximum transaction order Q for C .

Small Q

The decomposition gives a small transversal:

$$\text{lpt}(G) \leq 2Q + 1.$$

Double counting then gives a long path:

$$\ell + 1 \geq \frac{n}{2Q + 1}.$$

Large Q

Retain $\Omega(Q)$ connections, each of length $\Omega(Q)$.

The matching lemma combines many of them into one long path:

$$\ell = \Omega_{\beta}(Q^{1+\beta}).$$

Balancing the two bounds at $Q \asymp n^{1/(2+\beta)}$ gives

$$\ell = \Omega_{\beta}\left(n^{(1+\beta)/(2+\beta)}\right).$$

Bondy–Locke gives $c(G) \geq \frac{2}{5}\ell$; the cycle case is immediate.

Two approaches

Connected vertex-transitive G ($n \geq 3$); fix $0 < \beta \leq 1$.

Intersection approach

DeVos: counting

$$c(G) \geq \max\{\sqrt{Mn}, n/\text{lct}(G)\}$$

M : minimum intersection size of two longest cycles.

Strengthened Smith

$$\text{lct}(G) \leq S_1(M)$$

$$S_1(M) = O(M^{11/8})$$

Circumference

$$c(G) = \Omega(n^{19/30})$$

Transaction approach

Counting

$$\ell + 1 \geq n/\text{lpt}(G)$$

Quoted decomposition theorem

$$\text{lpt}(G) \leq 2Q + 1$$

Q is for the chosen cycle C .

Quoted matching lemma

Assume $\Omega_\beta(m^\beta)$ matching edges.

$$\implies \ell = \Omega_\beta(Q^{1+\beta})$$

Circumference

$$c(G) = \Omega_\beta(n/\ell^{1/(1+\beta)})$$

$$c(G) = \Omega_\beta(n^{(1+\beta)/(2+\beta)})$$

Cycle case immediate; otherwise $c(G) \geq 2\ell/5$ (Bondy–Locke). The right route is after reduction.

Thank you!

Questions

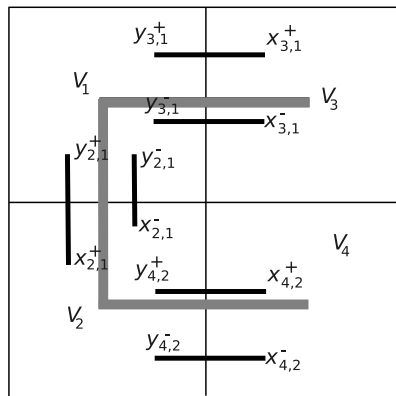
Length

Intersections

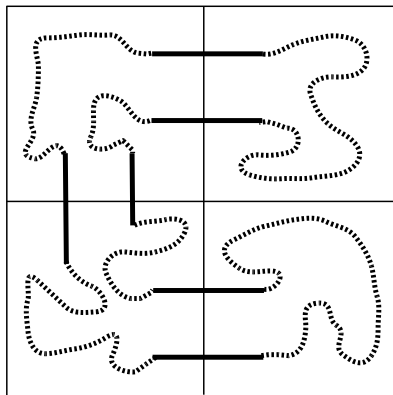
Transversals

Dense graphs: gluing

The case far from bipartiteness. The partition is already given.



(a) A tree on the pieces; two matching edges for each tree edge.



(b) Dotted paths cover each piece. The solid matching edges join them into one Hamiltonian cycle.